

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P_\theta$$

$$\hat{\theta}(X_1, \dots, X_n)$$

$$P(\cdot, \cdot)$$

$$P(\hat{\theta}, \theta)$$

$$\min_{\hat{\theta}} \mathbb{E}_{P_{\theta}}(P(\hat{\theta}, \theta))$$

$$\hat{\theta} = \theta$$

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \underbrace{\mathbb{E}_{P_{\theta}}(P(\hat{\theta}, \theta))}_{\text{lower-bound}} \geq m(\Theta, n, \dots)$$

① impossibility results

② Tightness of the upper bound

$$\mathbb{E}(\underbrace{\Phi(P(\hat{\theta}, \theta))}_{\text{increasing}})$$

Approach
→ a)

Markov
Bound

$$P(X \geq t) \leq \frac{\mathbb{E}(X)}{t}$$

b)



2δ - separated
Set

$$\{\theta^1, \dots, \theta^M\}$$

$$\forall i, j \in [M] \quad i \neq j \quad P(\theta^i, \theta^j) \geq 2\delta$$

Problem 1
 $x_1, \dots, x_n \stackrel{iid}{\sim} P_\theta$

$$x_1 \dots x_n \stackrel{iid}{\sim} P_\theta$$

$$\theta \equiv \hat{\theta}(x_1 \dots x_n)$$

➤

Problem 2

Problem 2

$x_1, \dots, x_n \stackrel{iid}{\sim} \frac{1}{M} \sum_{j=1}^M P_{\theta^j}(\cdot)$

$$J \equiv \hat{\theta}(x_1, \dots, x_n)$$



if I have an approx. of
the error here, it can act as
a lower bound for μ problem 1.
the

$$\text{let } X := \Phi(\rho(\hat{\theta}, \theta)) \geq 0 \xrightarrow{\text{Markov}} \mathbb{E}(\Phi(\rho(\hat{\theta}, \theta))) \geq \Phi(\delta) \cdot \mathbb{P}(\rho(\hat{\theta}, \theta) \geq \delta)$$

$\downarrow$

$\Phi(\rho(\hat{\theta}, \theta)) \geq \Phi(\delta)$

$$\Rightarrow \sup_{\theta \in \Theta} \mathbb{E}(\Phi(\rho(\hat{\theta}, \theta))) \geq \underbrace{\Phi(\delta) \cdot \mathbb{P}(\rho(\hat{\theta}, \theta) \geq \delta)}$$

Φ increasing

$$= \Phi(\delta) \sup_{\theta \in \Theta} \mathbb{P}(\rho(\hat{\theta}, \theta) \geq \delta)$$

$$\sup_{\theta \in \Theta} \mathbb{P}(\rho(\hat{\theta}, \theta) \geq \delta) \geq \frac{1}{M} \sum_{j=1}^M \mathbb{P}_{\theta^j}(\rho(\hat{\theta}, \theta^j) \geq \delta)$$

$\downarrow$

$\mathbb{Q}$ $\mathcal{Z} \sim \mathbb{P}_{\theta^j}(\cdot)$ $\mathcal{J} \sim \text{Unif}\{1, \dots, M\}$

$\mathbb{Q}(\mathcal{Z}) = \sum_{j=1}^M \underbrace{\mathbb{Q}(\mathcal{Z}, \mathcal{J})}_{\mathbb{P}_{\theta^j}} \underbrace{\mathbb{Q}(\mathcal{J})}_{1/M}$

$$\sup_{\theta \in \mathcal{L}} \mathbb{E}(\Phi(\rho(\hat{\theta}, \theta))) \geq \underbrace{Q(\rho(\hat{\theta}, \theta^J) \geq \delta)}_{\pi}$$

$\theta \in \mathcal{L}$

the red. from problem 1 to problem 2
has happened

$\hat{\theta}$ is a func. of Z .

$$\rightarrow \underbrace{\psi(Z)} = \arg \min_{j \in [M]} \rho(\underbrace{\hat{\theta}(Z)}, \theta^j)$$

an estimator

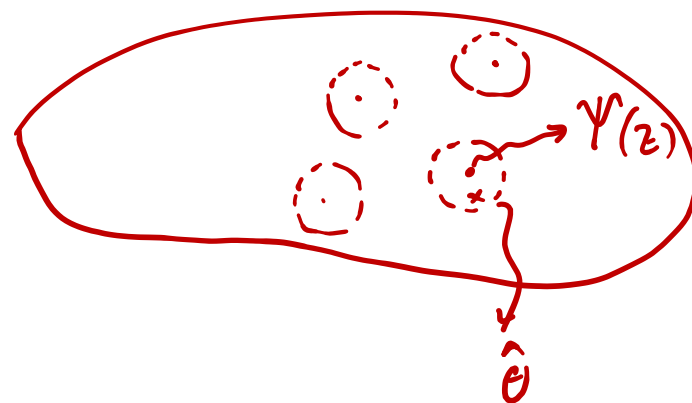
for the 2nd

problem using the

sol. that was originally

dev. for the 1st problem

①



Cond. on $J=j$

$$\{P(\hat{\theta}, \theta^j) < \underline{\delta}\} \Rightarrow \psi(z) = j$$

$$\theta^k \neq \theta^j$$

$k \in [m]$

$$P(\theta^k, \hat{\theta}) \geq \underbrace{P(\theta^k, \theta^j)}_{\geq 2\delta} - \underbrace{P(\theta^j, \hat{\theta})}_{\geq -\delta}$$
$$\geq \underline{\delta}$$

$$\Rightarrow \psi(z) = j$$

$$P(q_1) \leq P(q_2)$$

$$\text{or } P(\neg q_1) \geq P(\neg q_2)$$

$$Q(P(\hat{\theta}, \theta^J) \geq \delta) \geq Q(\psi(z) \neq J)$$

$$\Rightarrow \sup_{\theta \in \Theta} \mathbb{E}(\phi(\rho(\hat{\theta}, \theta))) \geq \phi(\delta) \cdot \underbrace{Q(\tilde{\psi}(z) \neq J)}_{\text{specific } \psi}$$

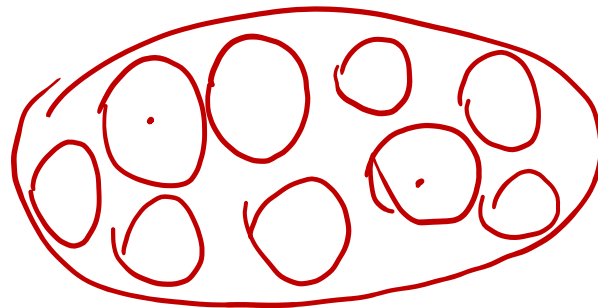
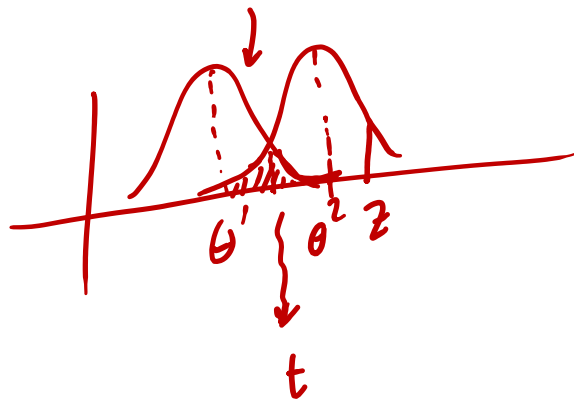
$\theta \in \Theta$

$$\geq \phi(\delta) \inf_{\tilde{\psi} \in \text{all func.} \Rightarrow \text{specific } \psi} Q(\psi(z) \neq J)$$

Best possible error for the 1st problem in worst case

$\tilde{\psi} \in \text{all func.} \Rightarrow \text{specific } \psi$

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}(\bar{\phi}(\rho(\hat{\theta}, \theta))) \geq \bar{\phi}(\delta) \cdot \inf_{\psi} Q(\psi(z) \neq J)$$



④ Opt. Hypothesis testing